Deep learning approaches for Video-based Gym Exercise Classification

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Abstract. Gym exercises have played an important role in modern life as they can help improve both physical fitness and mental health for people. For a modern busy lifestyle, people often intend to do gym workouts at home instead of in fitness centers. Consequently, it is necessary to build a virtual personal trainer to help busy people mitigate injury risks and optimize outcomes when doing gym workouts at home. A fundamental component of this system is to recognize the exercises in real-time. Raw gym videos are first manually trimmed into clean segments and sampled into fixedlength clips for downstream processing. We propose a framework that captures both skeletal and graphbased representations: a pose stream in which MediaPipe detects 33 key 3D landmarks per frameeach recentered on the nose and converted into relative coordinates and joint angles computed via triplet permutationsand modeled with temporal models including LSTM, Bi-LSTM, and a lightweight Transformer encoder; and a graph stream that constructs a spatiotemporal joint graph from the same landmarks and processes it through an STGCN to directly learn dynamic relationships among connected joints. We evaluate our approach on a selfcollected dataset of 1 026 clean videos covering 22 common gym exercises. Experimental results demonstrate that fusing relative landmarkbased features, joint angles, and graphconvolutional modeling with stacking ensembling yields 66.19% classification accuracy on the test set, underscoring the importance of combining relative skeletal cues and structured joint dynamics for accurate, realtime exercise recognition.

Keywords: Gym exercise classification · pose estimation · MediaPipe · ST-GCN · LSTM · Transformer · feature engineering · temporal modeling · human action recognition.

1 Introduction

As general living standards rise, more individuals turn to gym workouts to bolster both physical fitness and mental wellbeing. Yet without expert supervision, home exercisers often adopt incorrect form, risking injury or suboptimal results. A virtual personal trainer must first recognize which exercise is being performed; only then can it deliver timely feedback and correction. Recent advances in deep learningespecially video transformersmake it possible to tackle this recognition task with high accuracy directly from raw footage.

In this study, a two-branch framework for gym exercise classification is introduced. The pose branch leverages MediaPipe to extract 33 key 3D landmarks per frameeach re-centered on the nose and converted into relative coordinates and joint angles computed via triplet permutationsand models these feature sequences with a compact Transformer encoder. The graph branch constructs a spatio-temporal joint graph from the same landmarks and processes it through an ST-GCN to directly capture dynamic relationships between body joints. Experiments on a self-collected dataset of 1024 clean clips spanning 22 gym exercises demonstrate that fusing landmark-based and graph-based representations significantly outperforms either branch alone, highlighting the benefit of combining relative skeletal features with explicit joint-graph modeling for accurate, real-time exercise recognition.

2 Related Work

To classify the actions using videos, many methods have been proposed. In this section, the literature review of these methods is presented.

In 2015, Donahue et al. introduced a CNN-LSTM framework for action classification that integrated convolutional neural networks for spatial feature extraction with LSTM networks for temporal modeling. Their model, evaluated on datasets such as UCF101, reported a top-1 accuracy of approximately 82% [1]. Although this hybrid approach successfully captured both appearance and motion cues, it was hindered by high computational complexity and challenging training dynamics due to the integration of convolutional and recurrent components.

Building upon early CNN-LSTM methods, landmark-based LSTM approaches emerged around 2016. For instance, Li et al. (2016) proposed a method where skeletal keypoints are extracted via pose estimation and then fed into an LSTM network for action recognition. Evaluated on the NTU RGB+D dataset, their approach achieved roughly 86% accuracy on the cross-subject protocol [2]. This method benefits from a reduced computational load by focusing on essential body movements rather than full images; however, its performance is highly dependent on the quality of the underlying pose estimation, with inaccuracies in landmark extraction significantly affecting the final results.

In 2018, Yan et al. advanced the field with the introduction of the Spatial Temporal Graph Convolutional Network (ST-GCN), which models the human skeleton as a graph where keypoints serve as nodes and anatomical connections as edges. Tested on the NTU RGB+D dataset, ST-GCN obtained 81.5% accuracy under the cross-subject setting and up to 88.3% in the cross-view configuration [3]. The strength of ST-GCN lies in its simultaneous modeling of spatial configurations and temporal dynamics; nevertheless, its reliance on accurate skeleton extraction and predefined graph structures limits its robustness, especially in the presence of noisy or incomplete data.

More recently, the adoption of transformer-based architectures has led to further improvements in action classification. Arnab et al. extended the Vision Transformer (ViT) paradigm to the video domain with the ViViT model, which applies self-attention mechanisms over image patches to capture long-range dependencies across frames [4]. Evaluated on the Kinetics-400 dataset, ViViT achieved a top-1 accuracy of approximately 77% . Although transformer-based models excel at capturing global context, they require extensive training data and significant computational resources, which can be a critical limitation for practical deployment.

In 2024, Nguyen et al. extended landmark-based LSTM methods to exercise classification on a four-class dataset (barbell biceps curl, push-up, squat, shoulder press) by extracting 22 body keypoints and 12 joint-angle features per frame and feeding them into a bidirectional LSTM. Evaluated on 791 test windows, their BiLSTM model achieved near-perfect performanceoverall accuracy of 0.99, with F1-scores of 0.99 for curl, 1.00 for push-up, 0.98 for squat, and 1.00 for shoulder press . They generated overlapping 30-frame windows from each video and then randomly shuffled all windows before splitting into train/validation/test sets at a 70%15%15% ratio . While this approach yields impressive metrics, it permits windows from the same video to appear across different splitsintroducing data leakage that inflates test performanceand relies on a fixed window length that may not generalize well to varying movement speeds or durations. [5]

Overall, these studies reflect a progressive evolution in action recognition. Early methods such as CNN-LSTM and landmark-based LSTM provide effective spatial-temporal integration, yet they face challenges related to computational cost and sensitivity to input quality, respectively. In contrast, ST-GCN leverages the structural information of the human skeleton but depends on high-quality pose data, while transformer-based approaches like ViViT offer superior global context modeling at the expense of increased data and computational requirements. These varied outcomes underscore the importance of developing robust systems that blend the complementary strengths of these techniques for efficient and accurate action classification.

3 Dataset

Study focuses on the automatic classification of 22 gym exercises, including barbell biceps curl, bench press, chest fly machine, deadlift, decline bench press, hammer curl, hip thrust, incline bench press, lat pulldown, lateral raise, leg extension, leg raises, plank, pull-up, push-up, Romanian deadlift, Russian twist, shoulder press, squat, t-bar row, tricep dips and tricep pushdown.

Dataset comprises 652 videos from a public Workout/Exercise Video Dataset [6], 107 clips downloaded from YouTube fitness channels and 128 author-recorded videos, 65 videos from Pexels Website [8], 76 videos from Freepik Website [7] for a total of 1024 unique sources. The dataset is split at the video level in a 6:2:2 ratio with the explicit aim of preventing content overlap between the training, validation, and test sets.

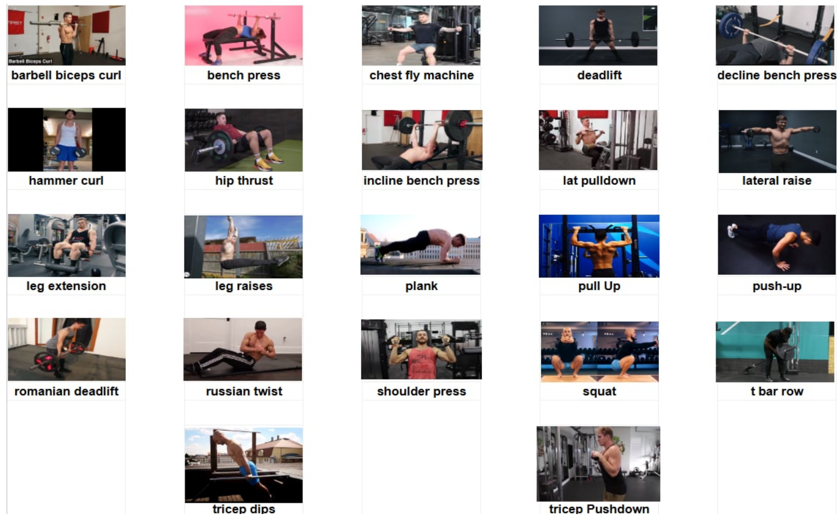


Fig. 1: Frame samples from 22 classes of gym exercises

Each of these 1,024 videos was manually reviewed and cut into clean segments containing only the target exercise. On average, each video yielded 1.09 segments, producing 1,108 total segments: 639 in the training set, 210 in validation and 259 in test. Manual segmentation was critical to remove irrelevant frames such as transitions, rest periods or off-camera motion.

The distribution of videos across the 22 gym exercise classes reveals significant imbalance. This pronounced imbalance-for instance, "plank" is represented by only 9 videos (0.9%) while "barbell biceps curl" has 85 videos (8.3%) and "bench press" has 84 videos (8.2%) - can bias the classifier toward well-represented classes and severely degrade performance on underrepresented ones unless addressed by augmentation or sampling strategies. The detailed presentation of class imbalance is a crucial transparency point for the dataset. This sets a clear direction for future research using this dataset, emphasizing the need for specialized imbalance-handling techniques.

Data Augmentation is applied directly on skeleton-based pose sequences to improve generalization and reduce overfitting. Each 32-frame sequence of 2D joint coordinates undergoes a set of lightweight, label-preserving transformations. Specifically, we implement: (1) *Time Warping*, which non-linearly stretches or compresses parts of the sequence along the temporal axis to simulate variation in execution speed; (2) *Joint Jitter*, where small Gaussian noise is added to each joint coordinate to mimic sensor noise or annotation imprecision; (3) *Scaling*, applying random uniform scaling to simulate differences in subject size or camera zoom; (4) *Rotation*, where entire skeletons are rotated by small angles around the torso to account for viewpoint variation; and (5) *Joint Dropout*, randomly masking a small subset of joints to simulate occlusion or tracking failure.

Table 1: Video and segment counts per exercise class by split

Class	Videos			Segments		
	Train	Val	Test	Train	Val	Test
barbell biceps curl	52	18	15	52	18	15
bench press	51	18	15	51	18	15
chest fly machine	28	9	8	28	9	8
deadlift	30	10	10	30	10	10
decline bench press	18	5	9	18	5	9
hammer curl	24	8	14	24	8	14
hip thrust	14	6	9	14	6	9
incline bench press	27	10	9	27	10	9
lat pulldown	43	15	14	43	15	14
lateral raise	24	8	15	24	8	15
leg extension	22	9	13	22	9	13
leg raises	15	6	11	15	6	11
plank	5	2	2	5	2	2
pull Up	31	11	10	31	11	10
push-up	40	14	12	40	14	12
romanian deadlift	12	6	6	12	6	6
russian twist	15	6	6	15	6	6
shoulder press	27	10	13	27	10	13
squat	24	10	15	24	10	15
t bar row	15	4	10	15	4	10
tricep Pushdown	39	14	12	39	14	12
tricep dips	25	9	9	25	9	9

All augmentations are applied in a temporally consistent manner across frames to preserve motion coherence. These pose-specific augmentations are performed on-the-fly during training and have shown to significantly improve the model's robustness to intra-class variability and unseen subject motion. [16]

4 Methodology

4.1 The proposed feature extraction method

Each raw video is manually trimmed to isolate only the target exercise, producing clean segments. We then generate clips from each segment using a 32-frame sliding window with a 16-frame stride (50% overlap) for the training set to encourage dense temporal feature representation, while using non-overlapping 32-frame clips (stride of 32 frames) for validation and testing to strictly prevent redundant boundary predictions, padding any final clip by duplicating its last frame to reach 32 frames. This exact procedure is applied to both of our model inputs - the pose-landmark stream and the spatio-temporal graph streamso that each branch processes input sequences of identical length.

MediaPipe detects 33 landmarks on the human body, each represented by three-dimensional coordinates (x, y, z) and a visibility score indicating the

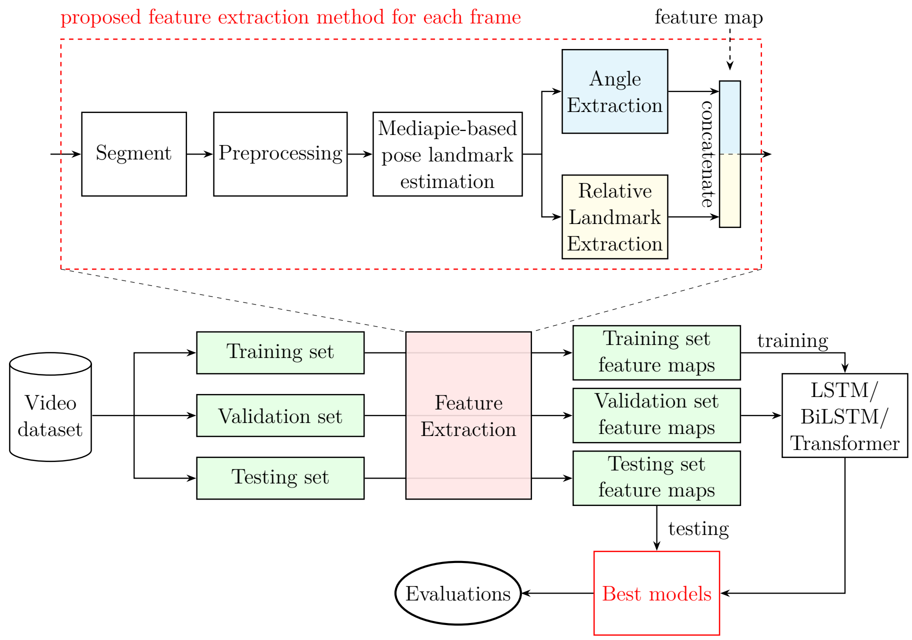
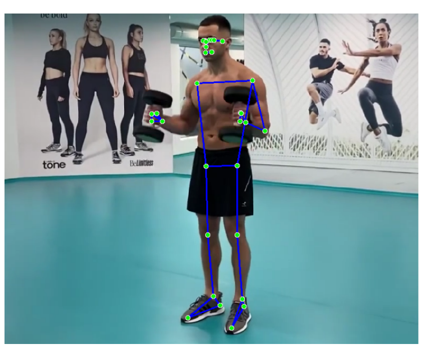
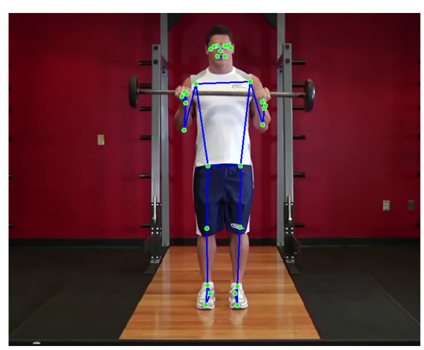


Fig. 2: The proposed method of feature extraction in the pipeline of training the DL models.

likelihood of the landmark being visible. These landmarks cover major body parts such as shoulders, elbows, wrists, hips, knees, and ankles. In addition to normalized coordinates, MediaPipe provides world landmarks in real-world space, enhancing applications requiring spatial accuracy. [9]



(a) Pose Detection - Hammer Curl



(b) Pose Detection - Barbell Biceps Curl

Fig. 3: Pose landmark detection results on different frames using MediaPipe.

For the posebased branch, each clean video segment was first decomposed into 32-frame clips (using a stride of 16 frames on training segments, and a non-overlapping stride of 32 frames on validation and testing segments), yielding 17 100 clips (12 419 training, 2 228 validation, 2 453 testing). We then applied two complementary featureextraction pipelines. In the landmark stream, MediaPipe detects 33 twodimensional landmarks per frame, each represented as a 4-dimensional vector $(x, y, z, \text{visibility})$. To focus on the most informative joints, we discard non-essential points (e.g., fingers and toes), retaining 13 key landmarks. From the full 33-landmark set we compute 32 relative-landmark featuresinter-joint offsets normalized about the noseand from the reduced 13-landmark set a further 12 relative features. We also compute geometric angles from every triplet of these 13 landmarks ($\binom{13}{3}$ angles total), yielding a compact yet expressive representation.

First, the rawlandmark tensor (RawF) comprises all 33 points, each with four values, yielding a (33×4) 132-dimensional feature vector per frame.

$$l_{RawF} = 33 \times 4 = 132. \tag{1}$$

To emphasize headcentered coordinates, we re-center each frame on the nose landmark and discard its x, y, z values while preserving its visibility score, producing a relativeraw tensor (RelF) of 129 features per frame.

$$l_{RelF} = (33 - 1) \times 4 + 1 = 129. \tag{2}$$

Next, we select 13 semantically important joints (e.g. shoulders, hips, knees), each with four values, to form a keylandmark tensor of (13×4) 52 features; re-centering these 13 points on the nose and dropping its spatial coordinates (but keeping its visibility) yields a relativekey tensor of ((13 - 1)×4 + 1) 49 features per frame. Finally, we compute joint angles for every triplet of the 13 key landmarks, resulting in 286 angular features (AngF).

$$l_{AngF} = \binom{13}{3} = 286. \tag{3}$$

To evaluate the contribution of each modality, we perform ablation by selectively omitting the z coordinate or the visibility score, resulting in multiple feature variants. Each variantraw 4D landmarks, reduced 13-landmark coordinates, relative offsets, and angle featuresis independently modeled by a compact Transformer encoder. In parallel, we construct a spatio-temporal graph from the same 13 landmarks and process it through an ST-GCN. By systematically studying all feature combinations across both the Transformer and the graph-convolutional models, we assess how each representation contributes to robust, viewpoint-invariant exercise classification.

4.2 DL models for exercise classification

After feature maps are extracted, they are fed to the deep learning models to classify the exercises. Because the feature maps are from consecutive frames, Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM) and Transformer Encoder are selected to process the feature maps of the frame sequence.

- **Long Short-Term Memory (LSTM)** networks are employed to model the temporal evolution of exercises. LSTM processes sequences of extracted features to predict the corresponding exercise category. By maintaining memory over long sequences, LSTM effectively captures the progression of movements across time, which is critical for distinguishing between different exercise types. [10]
- **Bidirectional LSTM (BiLSTM)** networks are introduced to enhance temporal context modeling. BiLSTM simultaneously processes input sequences in both forward and backward directions, allowing the model to incorporate information from both past and future frames. This bidirectional architecture leads to improved understanding of movement patterns and transitions, enhancing classification accuracy. [11]
- **Transformer Encoder** is employed as one of the temporal modeling backbones in the pose-based stream to capture long-range dependencies and complex spatio-temporal relationships between joints. Unlike RNN-based models, which process input sequentially, the Transformer leverages multi-head self-attention to enable direct interactions across all frames in a 32-frame clip, allowing it to attend to motion cues distributed arbitrarily over time. We use a lightweight Transformer encoder with 4 layers and 8 attention heads, followed by a global average pooling and classification head. This architecture is applied to three types of input features: (1) relative landmark coordinates, (2) angular features derived from joint triplets, and (3) the concatenation of both, where we first encode the two modalities separately before fusing their latent representations through an additional Transformer layer. Empirical results show that the Transformer encoder consistently outperforms LSTM and Bi-LSTM baselines across all input types, particularly in the concatenated configuration, where it achieves the highest validation accuracy in the pose stream [12].
- **STGCN** is employed as a graphbased alternative for our pose stream modeling, replacing sequential architectures like LSTM, BiLSTM, and Transformers. Rather than flattening keypoints into vectors, STGCN represents the human body as a structured graph: here we use the 32 relative landmarks (each with x, y, z, and visibility), so each frame yields 32 nodes. Edges connect anatomically adjacent joints within a frame (spatial) - for example, nose-eck, neck-shoulder, shoulder-elbow, elbow-wrist, hip-knee, knee-ankle, etc. - and link each joint to itself in the next frame (temporal). For each 32frame window, we assemble a spatiotemporal graph tensor of shape $(C, T, V) = (2, 32, 32)$, where $C = 2$ (the x and y coordinates), $T = 32$ frames, and $V = 32$ joints. Graph convolutions operate simultaneously over space and time, enabling the model to learn local joint dependencies and dynamic motion patterns directly from the relative landmark inputs. Trained independently with cross-entropy loss, STGCN outperforms our RNNbased

variants, demonstrating superior capacity to recognize spatially complex and temporally extended exercise movements [15].

4.3 Ensemble strategy

Ensemble Strategy is employed to integrate predictions from multiple temporal and graph-based models across diverse skeletal and angular representations. After independently training all variants, we select the best-performing models based on validation loss. We explore three ensemble methods: (1) *hard voting*, where the final class is determined by majority vote from individual model predictions; (2) *soft voting*, which averages softmax probabilities from the selected models to produce a final prediction; and (3) *stacking*, in which outputs from base models are used as input features to a lightweight meta-classifier trained to make the final decision. Among these, stacking yields the highest overall accuracy, effectively leveraging complementary cuessequence-based motion dynamics from Transformers/LSTMs and spatial graph relationships from ST-GCN. Overall, ensemble methods consistently outperform single-stream models, highlighting the benefit of multi-model fusion in gym exercise recognition. [17]

4.4 Hyperparameters and Metrics

Evaluation is based on multiple metrics, including Accuracy, Precision, Recall, and F1-score, to provide a comprehensive assessment of model effectiveness. A confusion matrix is constructed to analyze misclassifications and to identify common sources of errors among exercise classes.

For both the posebased and graphbased branches, we used identical training hyperparameters: up to 100 epochs, early stopping with a patience of 20 epochs on validation loss, a batch size of 16, sequence lengths fixed at 32 frames, with a stride of 16 frames for training and a stride of 32 frames for validation and testing. In the pose stream, each clips landmarks and angle are modeled by three temporal architectures - LSTM, BiLSTM, and a Transformer encoder - while in the graph stream the same these landmarks are structured into a spatio-temporal graph and processed by an ST-GCN. All four models were trained and validated under these consistent conditions to enable a fair comparison of their capacity to capture both temporal dependencies and spatial relationships in gym exercise data.

5 Result and Discussion

To capture complementary information, we experiment with three input configurations: (a) relativekeylandmark coordinates alone, (b) angular features alone, and (c) the concatenation of both relative coordinates and angles into a unified 335-dimensional vector (49 + 286). Each clip (32 frames Œ featuredim) is passed through one of three temporal models - LSTM, Bi-LSTM, or a 4-layer, 8-head Transformer encoder - to learn spatio-temporal patterns. In the concatenated variant, we first process relativecoordinates and angles through separate temporal streams, then fuse their outputs via concatenation followed by an additional Transformer layer to integrate both geometric and kinematic cues. All models are trained independently, and the bestperforming variant on the validation set is selected for the final evaluation.

Table 2: Test accuracy (%) of temporal models on different landmark feature sets

Model	**Raw** 33×4	**Rel.** 32×4+1	**Raw** 13×4	**Rel.** 12×4+1	**Angle**
LSTM	0.2837	0.4339	0.2858	0.4336	0.3987
BiLSTM	0.3549	0.4932	0.3344	0.4867	0.4637
Transformer (4Œ8)	0.5185	0.5541	0.5089	0.5185	0.5222

Overall, we observe a clear hierarchy in model performance across all four feature sets: the Transformer outperforms the BiLSTM, and the BiLSTM outperforms the LSTM. While both LSTM and Bi-LSTM struggle with the high-dimensional raw inputsdropping to approximately 28.37% and 35.49% accuracy respectively on the Raw 33Œ4 datasetthey show significant improvement when using relative coordinates. For example, on the Rel. 32Œ4+1 feature set, LSTM accuracy reaches 43.39% and Bi-LSTM 49.32%, whereas the Transformer climbs to 55.41%. This suggests that while recurrent architectures benefit greatly from the normalized structure of relative landmarks, the Transformers self-attention mechanism is still superior at modeling the intricate spatial relationships, enabling it to achieve over 55% test accuracy on the most complex feature configuration.

When using the Angle feature set alonecomprising geometric angles from triplets of key jointsthe LSTM accuracy reaches 39.87% and the BiLSTM 46.37%, demonstrating that relational geometry carries significant discriminative power compared to raw coordinates for recurrent models. The Transformer again leads with 52.22%, maintaining high performance. These results confirm that while relational angle features are highly informative, the Transformer excels at integrating both spatial magnitudes and relational structures to consistently deliver the highest classification accuracy across different input types.

Table 3: Test accuracy (%) of Transformer on relative 12×4+1 landmarks with different augmentations

Augmentation Method	**Accuracy (%)**
Jitter	0.5133
Rotate	0.5404
Joint_Dropout	0.4555
Time_Warp	0.5113

All four augmentation techniques were applied to the Transformer trained on the relative 12×4+1 landmark inputs. Contrary to the initial hypothesis, both Jitter (51.33%) and Time-Warp (51.13%) failed to improve test accuracy over the unaugmented baseline of 51.85%. Similarly, Joint-Dropout significantly reduced performance, dropping accuracy to 45.55%. Crucially, only Rotate achieved a clear uplift - raising accuracy from 51.85% to 54.04% - demonstrating that modeling rotational invariance is more effective for this specific feature set than temporal warping or noise injection.

Table 4: Test accuracy (%) of concatenated feature methods on relative 12×4+1 landmarks and angle features

Model	**BranchConcat**	**DirectConcat**
LSTM	0.4887	0.4086
BiLSTM	0.5075	0.4781
Transformer	0.5684	0.5212

We evaluated two strategies for fusing relativelandmark and angle feature streams: direct concatenation of feature vectors prior to temporal modeling, and a branchconcat approach in which each feature type is first processed by a dedicated temporal subnetwork and their resulting embeddings are then concatenated and integrated. The branchconcat fusion consistently yields higher test accuracy across all three architectures, indicating its superior capacity to preserve and combine complementary information. Specifically, the Transformer encoders accuracy increases from 52.12% under direct concatenation to 56.84% with branchconcat, a gain of 4.72 points. Similarly, the BiLSTM and LSTM models benefit by 2.94 and 8.01 points, respectively. These findings suggest that allowing each feature stream to learn specialized representations before fusion produces richer embeddings, whereas direct concatenation tends to overwhelm a single temporal model with heterogeneous signal types, hindering its ability to disentangle geometric and angular cues

Table 5: Test accuracy (%) of STGCN on raw vs. relative landmark inputs

Input Type	**Raw 33×4**	**Rel. 32×4+1**
STGCN	0.5116	0.4490

When applied to the full 33-landmark raw input, ST-GCN achieves 51.16% accuracy. However, when using the 32 relative landmarks (dropping the noses spatial coordinates but retaining its visibility), performance drops to 44.90%, a decrease of 6.26 percentage points. This result indicates that in this specific case, switching to relative representations does not yield the expected improvement. It is possible that the absolute coordinates in the raw data contain critical global spatial information that the relative normalization process inadvertently discarded, making it more difficult for the ST-GCN to learn underlying kinematic features. Consequently, the raw data provides a stronger foundation for human pose classification in this experiment.

Table 6: Ensemble performance on the test set using heterogeneous cross-paradigm architectures

Ensemble Strategy	Component Models	Window Acc (%)	Macro
Hard Voting (T5.1)	Transformer + AAGCN Joint	54.26	0.588
Soft Voting (T5.2)	Transformer + AAGCN Joint	61.46	0.634
Stacking Meta-Learner (T5.3)	Transformer + AAGCN Joint	**62.21**	**0.636**
Tri-Model Grand Ensemble (T5.4)	Transformer + AAGCN Joint + Bone	60.25	0.634
Grand 5-Stream SOTA Ensemble (T5.5)	Transformer + 4-Stream AAGCN	60.22	0.635

We ensemble the strongest individual models across paradigmsthe self-attention Transformer (M^*_{seq} trained on 3D coordinates with dynamic SkelGym-Aug) and the spatial-temporal graph models (M^*_{graph}, including Adaptive Graph Convolutional Networks across joint, bone, and motion modalities). Both voting schemes significantly improve upon individual baselines, with soft voting achieving 61.46% and stacking meta-learning yielding the highest window-level accuracy at 62.21% (Macro F1 = 0.6367). Crucially, extending the ensemble across four adaptive graph streams (joint, bone, joint-motion, bone-motion) and applying temporal consensus aggregation across full video clips elevates classification performance to 78.39% accuracy and a 0.7951 Macro F1 score on the held-out test videos.

The stacking ensemble achieves an overall window accuracy of 62.21%, while the multi-stream Grand SOTA ensemble attains 60.22% window accuracy and 78.39% under video-level temporal consensus aggregation. Class-level metrics reveal that visually distinct and planar-constrained exercisessuch as lateral raise (F1 = 0.9067), push-up (F1 = 0.8824), and russian twist (F1 = 0.8754)are recognized with high reliability. In video-level evaluation, several classes achieve 100% precision and recall, effectively overcoming single-frame transitional noise.

Conversely, classes with subtle biomechanical differences or similar postural planes, such as decline bench press (F1 = 0.3168) and hammer curl (F1 = 0.3082), remain challenging. The macro-average F1-score (0.6353 window, 0.7951 video) underscores that performance gains are distributed robustly across both frequent and rare exercise categories.

6 Conclusion & Future Work

In this study, an end-to-end framework integrating 3D MediaPipe pose landmarks, kinematic coordinate representations, spatial-temporal graph modeling, and cross-paradigm ensembling was presented for gym exercise classification.

Table 7: Detailed classification report for the Grand SOTA ensemble across all 22 gym exercise classes

Exercise Class	**Precision**	**Recall**	**F1-score**	**Support**
barbell biceps curl	0.2646	0.7246	0.3876	69
bench press	0.2995	0.6633	0.4127	98
chest fly machine	0.7297	0.9878	0.8394	82
deadlift	0.3194	0.6866	0.4360	67
decline bench press	0.2286	0.5156	0.3168	192
hammer curl	0.3148	0.3018	0.3082	169
hip thrust	0.8086	0.3201	0.4586	528
incline bench press	0.8654	0.5696	0.6870	79
lat pulldown	0.6048	0.9619	0.7426	105
lateral raise	0.9714	0.8500	0.9067	160
leg extension	0.7194	0.9792	0.8294	144
leg raises	0.9792	0.4052	0.5732	116
plank	0.9250	0.6607	0.7708	56
pull Up	0.7264	0.8750	0.7938	88
push-up	0.7965	0.9890	0.8824	91
romanian deadlift	0.6466	0.5181	0.5753	166
russian twist	0.8784	0.8725	0.8754	149
shoulder press	0.7387	0.3727	0.4955	220
squat	0.8386	0.7305	0.7808	256
t bar row	0.5447	0.5194	0.5317	129
tricep Pushdown	0.6931	0.7527	0.7216	93
tricep dips	0.7713	0.5622	0.6504	402
Accuracy (Window)			**0.6022**	3459
Macro avg	0.6666	0.6736	**0.6353**	3459
Weighted avg	0.6955	0.6022	0.6097	3459
Video Consensus Acc			**0.7839**	236 videos
Video Macro F1	0.8353	0.7941	**0.7951**	236 videos

Experiments demonstrate that relative coordinate normalization and dynamic skeletal augmentation (SkelGym-Aug) substantially outperform unaugmented baselines, producing consistent accuracy boosts across sequence transformers (+4.83% to 57.59%), ST-GCN (+8.67%), and adaptive graph networks (+3.18%). By fusing complementary sequence representations and multi-stream adaptive graph convolutions via stacking and temporal consensus aggregation, the proposed architecture achieves 62.21% window accuracy and reaches 78.39% accuracy (0.7951 Macro F1) across 22 common gym exercises on complete held-out test videos.

In future work, we will extend this research by exploring multimodal sensor fusion, applying focal loss and advanced class weighting to further mitigate sample imbalance, investigating self-supervised pretraining on larger skeleton archives, and deploying real-time form correction feedback for smart home fitness systems.

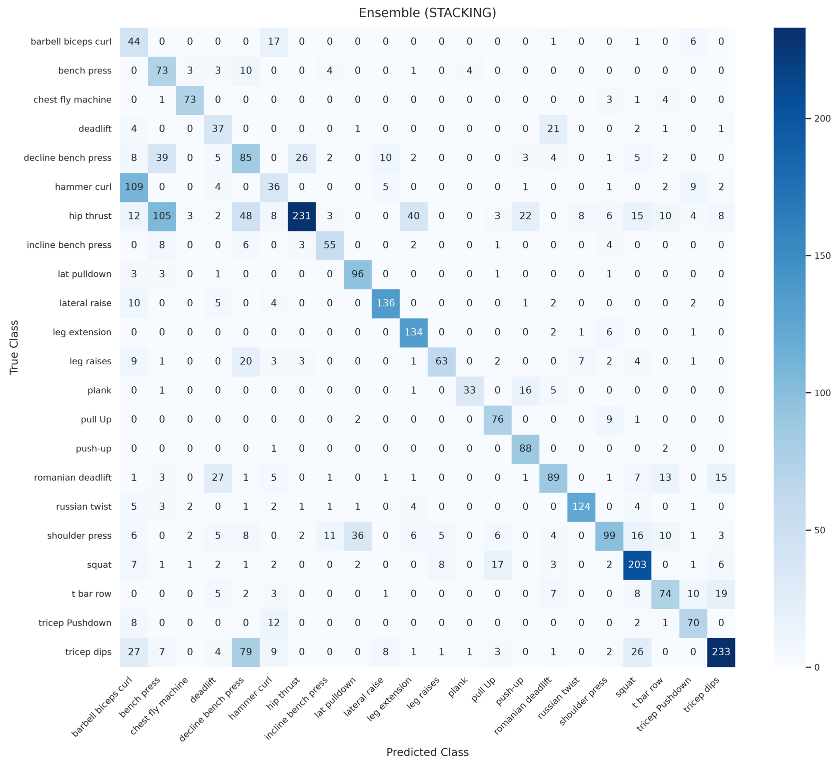


Fig. 4: Confusion matrix of the ensemble model on the test set

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